2.3.7.1. Control chart for a linear calibration line

Purpose

Line widths of three photomask reference standards (at the low, middle and high end of the calibration line) were measured on six days with an optical imaging system that had been calibrated from similar measurements on 10 reference artifacts. The control values and limits for the control chart , which depend on the intercept and slope of the linear calibration line, monitor the calibration and linearity of the optical imaging system.

Initial calibration experiment

The initial calibration experiment consisted of 40 measurements (not shown here) on 10 artifacts and produced a linear calibration line with:

- Intercept = 0.2357
- Slope = 0.9870
- Residual standard deviation = 0.06203 micrometers
- Degrees of freedom = 38

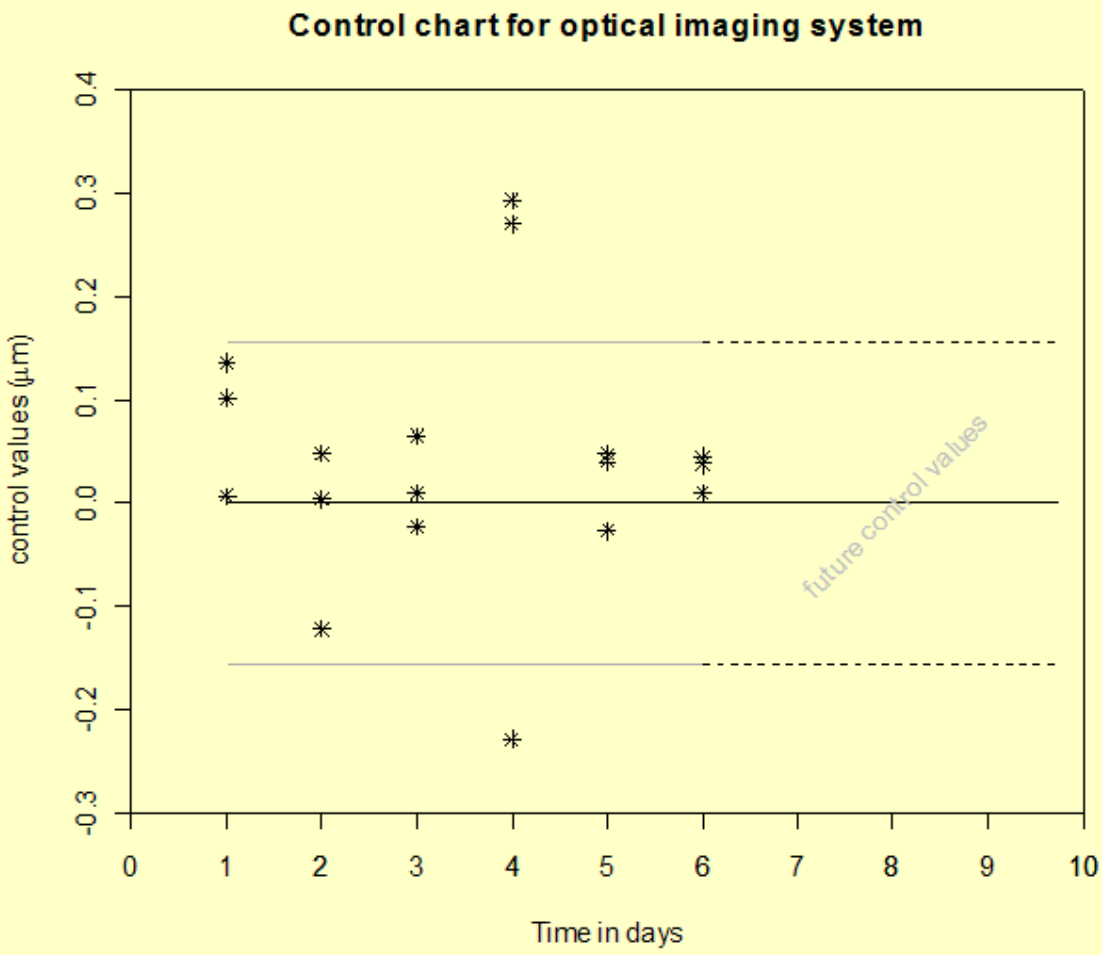
Line width measurements made with an optical imaging system

The control measurements, *Y*, and known values, *X*, for the three artifacts at the upper, mid-range, and lower end (U, M, L) of the calibration line are shown in the following table:

DAY	POSITION	X	Y
1	L	0.76	1.12
1	M	3.29	3.49
1	U	8.89	9.11
2	L	0.76	0.99
2	M	3.29	3.53
2	U	8.89	8.89
3	L	0.76	1.05
3	M	3.29	3.46
3	U	8.89	9.02
4	L	0.76	0.76
4	M	3.29	3.75
4	U	8.89	9.30
5	L	0.76	0.96
5	M	3.29	3.53
5	U	8.89	9.05
6	L	0.76	1.03
6	M	3.29	3.52
6	U	8.89	9.02

Control chart

The control chart shown below can be generated using both Dataplot code and R code. The reader can download the data as a text file.



Linewidths corrected for linear calibration
control values at lower, mid, and upper range of calibration interval

Interpretation of control chart

The control measurements show no evidence of drift and are within the control limits except on the fourth day when all three control values are outside the limits. The cause of the problem on that day cannot be diagnosed from the data at hand, but all measurements made on that day, including workload items, should be rejected and remeasured.

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